

Module 09 - Polls & Group Decisions

Real-time group voting for movies, restaurants, dates, destinations

SOURCE OF TRUTH: This resource intentionally uses plain JavaScript. The locked stack is React + JavaScript + Firebase + Material UI (MUI). Do not introduce Next.js, TypeScript, shadcn/ui, Redux, or another backend framework unless the project owner explicitly changes this specification.

Scope

Allow a plan/group organizer to create polls and participants to vote once per option set. Results update in realtime.

Poll document

```
groups/{groupId}/polls/{pollId}
{
  planId,
  question,
  status: "open" | "closed",
  allowMultipleChoices,
  createdBy,
  createdAt,
  closesAt
}
```

Option + vote

```
.../polls/{pollId}/options/{optionId}
{ label, order }

.../polls/{pollId}/votes/{uid}
{ selectedOptionIds, votedAt }
```

Rules

- One vote document per uid.
- If multiple choices are disabled, exactly one option is allowed.
- Closed poll rejects new votes.
- Only permitted organizers can close/delete polls.

UI

- Question header
- Option cards with vote counts
- Selected state
- Progress bars
- Close poll action for organizer
- Empty state when no options exist

Acceptance criteria

- Two clients see vote changes without refresh.
- Duplicate voting is prevented by data model/rules.
- Closing a poll is immediately reflected.

- Results are readable after closure.